5 Discussion

Our experiments put two very different ways of answering ALD/ALE questions side by side and make their strengths and weaknesses tangible. On the one hand, symbolic querying via SPARQL over ORKG comparisons is fast (microseconds per query on our hardware), deterministic (the same query over the same data always gives the same answer), and operates over an explicit schema of materials, processes, and performance descriptors. This matches how many JVSTA readers already think about process data: as rows and columns in a well-defined table or database.

On the other hand, neural querying over PDFs behaves very differently. When we ask LLMs to read the full review-article PDFs and generate table-style answers, queries take minutes rather than microseconds, results can change from run to run, and the models are prone to hallucinations, misread tables, and basic numerical errors—even when we provide carefully engineered natural-language versions of the SPARQL questions. In several cases, long chains of "step-by-step reasoning" cause the model to lose track of its own logic and drift into repetitive or partially inconsistent answers instead of converging on the precise comparison table we need.

Feeding LLMs the ORKG tables improves the situation but does not fully solve it. In the ORKG-augmented setting (Setting 3), we replace PDFs by compact, schema-aligned comparison tables exported as CSV. This clearly boosts answer quality and reduces some hallucinations: models move substantially closer to the SPARQL baseline and behave more predictably. However, they remain slower than direct SPARQL querying, still show some run-to-run variation, and still occasionally misinterpret column semantics or numeric values. In other words, structured tables make LLMs much more useful, but they do not turn them into perfect drop-in replacements for symbolic querying.

These observations underline that our main contribution is to make the case for *keeping* a symbolic layer for scientific knowledge, rather than claiming that we have built a full neurosymbolic system. Our Setting 3 is best understood as symbolically grounded neural question answering: the ORKG comparison (a structured, human-curated table) constrains and guides a language model that remains fundamentally stochastic, i.e., it samples from a probability distribution over possible next tokens in the generated sequence, so that the same question can yield slightly different outputs from run to run. We do not change the model architecture or training objective, and we do not integrate a logical reasoning module inside the network. Instead, we show empirically that even the strongest LLMs should not be treated as substitutes for a symbolic query engine in deterministic scientific applications. At their best, they approximate the outputs of SPARQL and benefit dramatically from being *fed* symbolic structure that has been curated and maintained in a knowledge graph such as the ORKG.

Looking ahead, our results suggest that symbolic question answering and LLM-based question answering should be viewed as *complementary* for ALD/ALE research. Symbolic querying over ORKG via SPARQL is strongest when the goal is exact, repeatable filtering and aggregation of well-defined quantities (e.g., "which ALD recipes for material X satisfy a given temperature window and growth-per-cycle constraint?"). In these cases, every row and every number matters, and the symbolic layer provides hard guarantees: the same query over the same data always returns the same result, with transparent provenance. LLM-based question answering, by contrast, is at its best for more open-ended, exploratory, or integrative tasks: explaining mechanisms, relating processes to broader trends, or combining scattered observations into a coherent narrative. This view is consistent with recent ALD benchmarking work, which evaluates GPT-4-class models as assistants for conceptual and design questions rather than as exact table-reproducing engines [65], and with MOF-ChemUnity, which couples a literature-derived knowledge base for metal–organic frameworks with LLM-driven assistance for retrieval and structure–property reasoning [53]. In both cases, LLMs are most convincing when they sit on top of curated knowledge, not when they are the sole source of quantitative truth.

On the symbolic side, our work also connects to large, literature-derived knowledge graphs in materials science such as MatKG [61], which autonomously extracts entities and relations from millions of materials papers and exposes them as an RDF/CSV knowledge graph queryable via SPARQL. While MatKG emphasises broad, automatically mined coverage, our ORKG-based comparisons focus on a smaller set of review tables with explicit schema design, expert curation, and process-level detail tailored to ALD/ALE. Together, these efforts illustrate a spectrum of symbolic infrastructures: from wide, automatically generated graphs to high-fidelity, machine-actionable review tables that can serve as gold-standard targets for precise querying and future neurosymbolic integration.

In light of this, a natural next step is to explore *closer integrations of symbolic knowledge with neural models*, given the case for symbolic knowledge made in this paper. One direction would be to train or fine-tune LLMs directly on symbolic artefacts—comparison tables, ORKG triples, and SPARQL query–result pairs—so that they become better at precise, table-like operations while still supporting trend extrapolation and hypothesis generation (for example, suggesting candidate precursors, process windows, or follow-up experiments). For such approaches to be effective, however, the underlying symbolic data must be both *verified* and *large-scale*. Since current LLMs are already trained on massive amounts of natural language text, the bottleneck shifts to constructing sufficiently rich gold-standard symbolic datasets. This calls for workflows that can mine machine-actionable knowledge from large collections of papers with very high accuracy (ideally well above 95% for table reconstruction and key fields), combining automatic extraction with targeted human validation. Prospectively, we therefore invite the ALD/ALE community to publish machine-actionable versions of their review tables—alongside the traditional PDF—as part of a growing gold-standard set (for example in ORKG and linked repositories). Another complementary direction is to design hybrid systems in which a deterministic symbolic core (e.g., a SPARQL engine over ORKG) remains responsible for exact filtering and aggregation, while neural components help with formulating natural-language queries, ranking and clustering results, and explaining patterns. Across these scenarios, our findings point to a clear design principle for ALD/ALE workflows: reliable, quantitative querying should remain anchored in symbolic substrates such as ORKG, with neural models used as powerful but approximate assistive layers on top—well suited for exploration, explanation, and idea generation, but not treated as standalone sources of truth for process-critical numbers.

6 Conclusion

This work showed how ALD/ALE review knowledge can be transformed from static PDF tables into a machine-actionable layer that supports precise, repeatable querying. From nine recent review articles, we converted 18 multi-study process tables into ORKG *comparisons* and defined 33 natural-language questions with corresponding SPARQL queries, releasing them as the ALD/E Query Dataset [15]. A survey with three ALD/ALE experts confirmed that these machine-actionable reviews are scientifically meaningful and practically useful for tasks such as precursor screening, process-window comparison, and mechanism overview, while also highlighting concrete opportunities to enrich schemas and add missing parameters. Comparing three ways of answering the same questions—symbolic querying via SPARQL over ORKG, purely neural querying over review PDFs, and neural querying grounded in ORKG comparison tables—shows a clear division of roles: SPARQL over ORKG remains the gold standard for exact, reproducible answers with transparent provenance, whereas large language models are best used as assistive layers on top, especially when grounded in structured ORKG tables, for formulating questions, exploring patterns, and generating hypotheses.

Looking ahead, our results argue for growing and tightening the connection between symbolic and neural tools. On the symbolic side, we see a need for community workflows that routinely publish machine-actionable versions of review tables alongside PDFs, creating large, verified corpora that can serve both as gold-standard resources and as training material for future neurosymbolic models. On the neural side, hybrid systems in which a deterministic symbolic core remains responsible for exact filtering and aggregation, while LLMs handle query formulation, result exploration, and explanation, appear particularly promising. Finally, this work complements emerging efforts on multimodal scientific AI, where models are expected to interpret not only text and tables but also the figures that carry much of the quantitative insight in ALD/ALE and related fields [14]. Taken together, these directions outline a path toward scientific assistants that can reliably answer precise questions, help make sense of complex datasets, and eventually reason across text, tables, and images in an integrated way.

Acknowledgments and Disclosure of Funding

Experiments with the five open-weights LLMs were supported by the KISSKI HPC infrastructure [13], in particular via the ChatAI API service (`https://kisski.gwdg.de/en/leistungen/2-02-llm-service/`). This centralized service provides access to very large language models

with billions of parameters, whose computational demands often exceed the local GPU resources typically available in academic research environments, and thereby alleviates the need for redundant deployments and time-consuming ad hoc querying setups.

This work is supported by the program “AI-Aware Pathways to Sustainable Semiconductor Process and Manufacturing Technologies”, co-sponsored by Intel Corporation & Merck KGaA, Darmstadt, Germany.

Data Availability Statement

The code accompanying this article is released on Github at `https://github.com/sciknoworg/ald-ale-orkg-review` with detailed documentation. The dataset produced and studied in this research is released on Zenodo [15] and be accessed at this DOI `https://doi.org/10.5281/zenodo.17720429`.

References

[1] Amr Ghazy, David Zanders, Anjana Devi, and Maarit Karppinen. Rare-earth dopants in ald thin films: Host materials, dopants, and applications, 2025. URL `https://orkg.org/comparison/R1469955`.

[2] Amr Ghazy, David Zanders, Anjana Devi, and Maarit Karppinen. Atomic layer deposition processes of rare-earth-based thin films, 2025. URL `https://orkg.org/comparison/R1471077`.

[3] Amr Ghazy, David Zanders, Anjana Devi, and Maarit Karppinen. Rare-earth ald/mld processes for r–organic hybrid thin films, 2025. URL `https://orkg.org/comparison/R1470110`.

[4] Andreas Fischer, Aaron Routzahn, Steven M. George, and Thorsten Lill. Summary of reported thermal atomic layer etching processes by material and reactants, 2025. URL `https://orkg.org/comparison/R1563034`.

[5] Anthropic. Introducing claude sonnet 4.5. `https://www.anthropic.com/news/claude-sonnet-4-5`, September 2025. Accessed: 2025-11-21.

[6] Sören Auer, Allard Oelen, Muhammad Haris, Markus Stocker, Jennifer D’Souza, Kheir Eddine Farfar, Lars Vogt, Manuel Prinz, Vitalis Wiens, and Mohamad Yaser Jaradeh. Improving access to scientific literature with knowledge graphs. *Bibliothek Forschung und Praxis*, 44(3):516–529, 2020.

[7] Sören Auer, Jennifer D’Souza, Kheir Eddine Farfar, Mohamad Yaser Jaradeh, Azanzi Jiomekong, Oliver Karras, Allard Oelen, Lauren Snyder, Markus Stocker, and Lars Vogt. Open research knowledge graph: a large-scale neuro-symbolic knowledge organization system. In *Handbook on Neurosymbolic AI and Knowledge Graphs*, pages 385–420. IOS Press, 2025.

[8] Caio Viktor S Avila, Vânia MP Vidal, Wellington Franco, and Marco A Casanova. Experiments with text-to-sparql based on chatgpt. In *2024 IEEE 18th international conference on semantic computing (ICSC)*, pages 277–284. IEEE, 2024.

[9] Vannevar Bush et al. As we may think. *The atlantic monthly*, 176(1):101–108, 1945.

[10] Chang Fang, Yanqiang Cao, Di Wu, and Aidong Li. Typical thermal atomic layer etching (ale) processes for representative materials, 2025. URL `https://orkg.org/comparison/R1560222`.

[11] Marco Cremaschi, Federico Belotti, Jennifer D’Souza, and Matteo Palmonari. Mammotab 25: A large-scale dataset for semantic table interpretation-training, testing, and detecting weaknesses. In *International Semantic Web Conference*, pages 131–148. Springer, 2025.

[12] Doo San Kim, Ju Eun Kim, You Jung Gill, Yun Jong Jang, Ye Eun Kim, Kyong Nam Kim, Geun Young Yeom, and Dong Woo Kim. Summary of reported anisotropic and isotropic atomic layer etching (ale) processes for metals, 2025. URL `https://orkg.org/comparison/R1563131`.